1 part b.)

Using approach for

$A_2X = b$

$$b = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

* $Ly = b$
  $Ux = y$

$[L \quad b] \longrightarrow [I \quad y]$

$[U \quad y] \longleftarrow [I \quad x]$

$$L_2 = \begin{pmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 0 & -1/2 & 1 \end{pmatrix}$$

$$A_2 \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$$U_2 \begin{pmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1/2 \\ 0 & 0 & 4/3 \end{pmatrix}$$

(L1b)

$$= \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ -1/2 & 1 & 0 & 2 \\ 0 & -1/2 & 1 & 3 \end{array}\right)$$

$\xrightarrow{2R2 + R1 \to R2}$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 5 \\ 0 & -1/2 & 1 & 3 \end{array}\right)$$

$\xrightarrow{\frac{1}{4}R3 + R2 \to R2}$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 5 \\ 0 & 0 & 4 & 17 \end{array}\right)$$

$\xrightarrow{\begin{array}{l} 1/2R2 \to R2 \\ 1/4R3 \to R3 \end{array}}$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 5/2 \\ 0 & 0 & 1 & 17/4 \end{array}\right) = (I \mid \vec{y})$$

$$\vec{y} = \begin{pmatrix} 1 \\ 5/2 \\ 17/4 \end{pmatrix}$$

$\frac{262}{96} + \frac{96}{96} \quad \frac{51}{32} + \frac{64}{32}$

$[U|\vec{y}] =$ $\quad \cdot \frac{4}{3}\frac{3}{8} = \frac{1}{2}$

$$\left(\begin{array}{ccc|c} 2 & -1 & 0 & 1 \\ 0 & 3/2 & -1/2 & 5/2 \\ 0 & 0 & 4/3 & 17/4 \end{array}\right) \xrightarrow{\frac{3}{8}R3 + R2 \to R2}$$

$$\left(\begin{array}{ccc|c} 2 & -1 & 0 & 1 \\ 0 & 3/2 & 0 & 131/32 \\ 0 & 0 & 4/3 & 17/4 \end{array}\right) \xrightarrow{\frac{2}{3}R2 + R1 \to R1}$$

$$\left(\begin{array}{ccc|c} 2 & 0 & 0 & 358/96 \\ 0 & 3/2 & 0 & 115/32 \\ 0 & 0 & 4/3 & 17/4 \end{array}\right)$$

$\frac{1}{2}R1 \to R1$

$\frac{2}{3}R2 \to R2$

$\frac{3}{4}R3 \to R3$

$$\vec{X} = \begin{pmatrix} 358/96 \\ 115/32 \\ 17/4 \end{pmatrix}$$